

Real-Time Pedestrian Attribute Recognition with YOLOv8 and ResNet18

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Abstract

Pedestrian attribute recognition is critical for intelligent surveillance and human-centric analysis. This paper presents an integrated deep learning framework combining YOLOv8 for real-time pedestrian detection and ResNet18 for attribute classification. The system supports gender classification, age estimation, and multi-attribute recognition across 61 binary attributes including clothing, colors, and accessories.

A novel contribution is the creation of a unified dataset schema merging PETA and PA-100K through attribute-space mapping, resulting in over 100,000 training images. Transfer learning and class imbalance strategies further improve robustness.

Experiments demonstrate near-perfect gender classification (99.89% accuracy). The multi-attribute system achieves 95.56% overall accuracy with a micro-average F1-score of 0.514. Key attributes including "LowerBody-Trousers" (0.811 F1), "UpperBody-ShortSleeve" (0.687 F1), and "Carrying-MessengerBag" (0.519 F1) show strong performance. The system is deployed as a web-based solution with Flask/SocketIO for real-time communication. Results confirm robust performance across varying illumination, viewpoints, and poses, providing a scalable foundation for smart city surveillance.

Keywords: Pedestrian attribute recognition, YOLOv8, ResNet18, deep learning, computer vision

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1. Introduction

Pedestrian Attribute Recognition (PAR) is a fundamental yet challenging task in computer vision, with applications spanning intelligent surveillance, transportation systems, and human-computer interaction [1, 2, 3]. The ability to automatically infer demographic and appearance-based descriptors from individuals in visual data unlocks transformative applications across numerous sectors, including public security and forensic search, intelligent transportation systems, smart retail environments, and next-generation human-computer interaction systems [4, 5].

Historically, surveillance systems have been predominantly passive and reliant on continuous human oversight. Human operators monitor multiple video feeds—a process that is economically burdensome and limited by cognitive constraints such as attention fatigue and subjective bias [6]. In the era of massive-scale urban surveillance with millions of camera units generating petabytes of data daily, this model has become unsustainable. The shift toward automated, intelligent vision systems is therefore a strategic necessity. Automated PAR systems process vast streams of visual information in real-time, converting unstructured pixel data into structured, searchable metadata, effectively transforming video archives into queryable databases of human attributes [7, 8].

The paradigm shift enabling this automation is largely attributable to deep learning, particularly Convolutional Neural Networks (CNNs). Since AlexNet’s landmark success in 2012 [9], CNN architectures have undergone rapid evolution, with models such as VGG [10], GoogLeNet [11], ResNet [12], DenseNet [13], EfficientNet [14], and MobileNet [15] consistently surpassing performance benchmarks. A pivotal advantage is robust transfer learning, allowing researchers to leverage models pre-trained on large-scale datasets like ImageNet [9] and fine-tune them for specific, data-scarce domains such as PAR [16].

Despite these powerful tools, PAR faces unique challenges. First, in-the-wild imaging conditions present significant hurdles: pedestrians appear at vastly different scales, face occlusions, and are imaged under diverse lighting conditions from arbitrary viewpoints [17, 18]. Second, PAR is inherently a multi-label classification task—a pedestrian simultaneously possesses multiple attributes (e.g., male, 30 years old, wearing a jacket and jeans, carrying a backpack) that are not mutually exclusive [19, 20]. Third, PAR requires a two-stage pipeline: accurate pedestrian detection and localization as a pre-

requisite, followed by fine-grained attribute classification on tightly cropped regions. Failure in the first stage directly leads to failure in the second [21]. Fourth, severe class imbalance is pervasive—attributes like "wearing a hat" appear infrequently compared to "wearing trousers," causing models to bias toward majority attributes and perform poorly on rare but potentially critical ones [22, 23].

1.1. Problem Statement

Despite significant progress in pedestrian detection and attribute recognition, existing systems face several limitations [16, 5]. First, most approaches focus on either detection or attribute recognition separately, lacking an integrated end-to-end solution. Second, many systems are trained on single datasets, limiting generalization to diverse real-world scenarios [2]. Third, the trade-off between accuracy and inference speed remains critical, as many high-accuracy models are too slow for real-time applications [24].

The primary research problem addressed in this paper is: *How can we design and implement an efficient, accurate, and practical pedestrian attribute recognition system that operates in real-time while maintaining high accuracy across diverse scenarios?*

Specifically, this research addresses the following sub-questions:

- How can pedestrian detection and attribute recognition be effectively integrated into a unified system?
- What is the optimal architecture for simultaneously recognizing multiple pedestrian attributes including gender, age, and appearance characteristics?
- How can multiple datasets be leveraged to improve model generalization and robustness?
- What training strategies and loss functions are most effective for multi-attribute recognition tasks?
- How can the system be optimized for real-time performance while maintaining high accuracy?

1.2. Contributions

The main contribution of this paper is the design and implementation of a comprehensive pedestrian attribute recognition system that achieves high accuracy and real-time performance [25, 12]. The specific contributions are:

- **Detection Framework:** An accurate and fast pedestrian detection system using YOLOv8 that reliably detects pedestrians under various conditions including different scales, lighting, and occlusions [26, 27].
- **Attribute Classification:** ResNet18-based models for binary gender classification, continuous age estimation, and multi-attribute recognition covering 61 attributes including clothing types, colors, accessories, and carried items [12, 11].
- **Dataset Integration:** A combined dataset strategy merging PETA and PA-100K datasets to improve model generalization through attribute-space mapping [28, 29].
- **End-to-End System:** A complete system supporting multiple input modalities (images, videos, live camera feeds) with a web-based interface [30].
- **Real-Time Optimization:** Optimizations for real-time processing targeting speeds suitable for practical surveillance applications [20, 6].

1.3. Scope and Limitations

This research focuses on pedestrian attribute recognition using deep learning methods, encompassing detection using YOLOv8, gender classification, age estimation, multi-attribute recognition covering 61 attributes, training on PETA [28] and PA-100K [29] datasets, and a web-based user interface.

Several limitations should be acknowledged. The system is primarily evaluated on publicly available datasets (PETA and PA-100K) and may require fine-tuning for specific deployment scenarios [23]. Age estimation accuracy may vary for certain age groups due to inherent biases in training data [31]. The system focuses on frontal and side views typical in surveillance scenarios; extreme camera angles or heavy occlusions may reduce accuracy [6]. Real-time performance depends on hardware capabilities—the system is optimized for modern GPUs, but performance may be reduced on CPU-only systems. Privacy and ethical considerations, while acknowledged as important, are not the primary focus of this technical research [22].

1.4. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work in pedestrian detection, attribute recognition, and relevant datasets. Section 3 describes the system architecture, dataset preparation, model designs, and training methodology. Section 4 presents the experimental setup and results for gender classification, age estimation, and multi-attribute recognition. Section 5 concludes with a summary of contributions, limitations, and future work.

2. Related Work

This section reviews existing research in pedestrian detection, attribute recognition, and relevant datasets that form the foundation of this work [2, 3].

2.1. Pedestrian Detection Methods

Pedestrian detection has been a fundamental problem in computer vision for decades [32]. Early approaches relied on handcrafted features such as Histogram of Oriented Gradients (HOG) combined with Support Vector Machine (SVM) classifiers. The HOG descriptor, proposed by Dalal and Triggs [32], became a standard feature extraction method due to its ability to capture local shape information. However, these traditional methods struggled with variations in scale, pose, and lighting conditions [27].

The introduction of deep learning revolutionized pedestrian detection. Region-based CNN (R-CNN) methods [33] achieved significant improvements by using convolutional neural networks to extract features from region proposals. However, R-CNN and its variants (Fast R-CNN, Faster R-CNN) [34] were computationally expensive and not suitable for real-time applications [27].

The development of single-stage detectors addressed the speed limitations of two-stage methods. YOLO (You Only Look Once) [25] revolutionized real-time object detection by treating detection as a single regression problem. YOLO divides the input image into a grid and predicts bounding boxes and class probabilities directly from full images in one evaluation [25]. Subsequent versions (YOLOv2 through YOLOv8) have continuously improved accuracy and speed [26].

YOLOv8, released by Ultralytics in 2023 [26], represents the latest advancement in the YOLO series. It introduces architectural improvements including a new backbone network, an anchor-free detection head, and improved training strategies. YOLOv8 achieves state-of-the-art performance

on the COCO dataset while maintaining real-time inference speeds, making it ideal for pedestrian detection in surveillance applications [25, 27]. For our work, we select YOLOv8n (nano variant) as the detection backbone due to its balance between accuracy and speed.

Other notable detection architectures include SSD (Single Shot MultiBox Detector) [27], RetinaNet with focal loss [35], and transformer-based detectors like DETR [36]. However, YOLOv8 offers the best trade-off for our real-time requirements.

2.2. Attribute Recognition Approaches

Pedestrian attribute recognition has evolved from simple binary classification to complex multi-label learning problems [29, 28]. Early approaches treated attribute recognition as independent binary classification tasks, training separate classifiers for each attribute [37]. However, this approach ignores correlations between attributes and is computationally inefficient [38].

Deep learning approaches have shown remarkable success in multi-attribute recognition [39, 38]. Convolutional Neural Networks (CNNs) learn hierarchical feature representations that capture both low-level visual features and high-level semantic information. ResNet architectures [12] enable training of very deep networks through residual connections, addressing the degradation problem in deep networks [11].

Transfer learning has been crucial for attribute recognition tasks, allowing researchers to leverage knowledge from large-scale datasets like ImageNet [9]. Pre-trained CNN models can be fine-tuned for specific attribute recognition tasks with relatively small amounts of task-specific data [10]. This approach has been successfully applied to pedestrian attribute recognition, achieving state-of-the-art results on benchmark datasets [29, 28].

Recent research has explored attention mechanisms [39, 3], part-based models [19], multi-task learning frameworks [38, 4], and imbalanced classification strategies [40, 35]. Attention mechanisms help models focus on relevant image regions for each attribute [1], while part-based models explicitly model different body parts [19]. Multi-task learning frameworks jointly optimize multiple related tasks, potentially improving performance through shared representations [38, 7].

More recent works have explored graph-based methods [2], transformer architectures [16], and cross-dataset learning [21]. However, these methods often come with higher computational costs, making them less suitable for real-time deployment.

This paper adopts a ResNet18-based architecture with transfer learning for attribute recognition. The approach uses separate models for gender classification, age estimation, and multi-attribute recognition, allowing task-specific optimization while maintaining computational efficiency [12, 30].

2.3. Datasets for Pedestrian Attribute Recognition

Several large-scale datasets have been created for pedestrian attribute recognition research [2].

PETA (People in Photo Albums) [28] contains 19,000 pedestrian images with annotations for 61 binary attributes and 4 multi-class attributes. The dataset includes diverse scenarios with variations in pose, scale, and lighting conditions, and has become a standard benchmark for PAR research [29].

PA-100K (Pedestrian Attribute 100K) [29] contains 100,000 pedestrian images with annotations for 26 attributes. The dataset focuses on surveillance scenarios and includes diverse pedestrian appearances with pre-defined train/val/test splits [29].

Market-1501 and **DukeMTMC** [41, 42] are primarily designed for person re-identification but include attribute annotations, providing additional benchmarks for attribute recognition [43].

RAP (Richly Annotated Pedestrian) dataset contains 41,585 images with 72 attributes, providing fine-grained annotations for detailed attribute recognition research [29].

UPAR (University Pedestrian Attribute Recognition) [6] and other recent datasets have further expanded the available data for PAR research.

This paper primarily uses PETA and PA-100K datasets, as they provide comprehensive attribute annotations and are widely recognized benchmarks [28, 29]. A key contribution of our work is the creation of a combined dataset that merges PETA and PA-100K, mapping PA-100K attributes to PETA’s attribute space to create a larger, more diverse training set [21, 23].

2.4. Summary of Related Work

In summary, existing research has established strong foundations in pedestrian detection (from HOG-SVM to YOLO and transformers) and attribute recognition (from single-task to multi-label learning with transfer learning, attention, and graph-based methods) [2, 3]. However, most systems treat detection and recognition as separate problems, and many are trained on single

datasets [5]. Our work addresses these gaps by: (1) integrating YOLOv8 detection with ResNet18-based attribute recognition in an end-to-end pipeline, (2) combining multiple datasets (PETA and PA-100K) through attribute-space mapping, and (3) optimizing for real-time performance suitable for practical surveillance applications [16, 22].

3. Methodology

This section describes the overall system architecture, dataset preparation, model designs, and training methodology of our pedestrian attribute recognition framework.

3.1. System Architecture Overview

The proposed system follows a two-stage pipeline, as illustrated in Figure 1. First, YOLOv8n [26] detects pedestrians and produces bounding boxes. Second, the detected pedestrian regions are cropped, resized to 224×224 , normalized, and passed to ResNet18 [12] networks for gender classification, age estimation, and multi-attribute recognition [25].

The system supports three input modalities: static images, video files, and live camera feeds. The modular design allows future improvements such as replacing the detection model or adding new attribute categories [30].

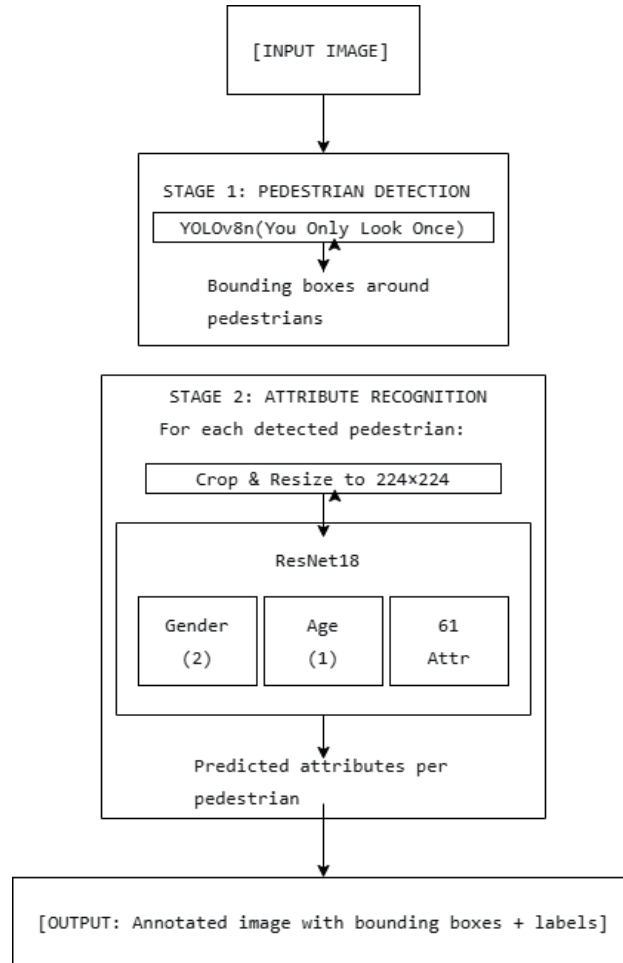


Figure 1: Two-stage pipeline of the proposed pedestrian attribute recognition system. Stage 1: YOLOv8 detects pedestrians. Stage 2: ResNet18 predicts gender, age, and 61 attributes for each detected pedestrian.

Table 1: Main system components

Component	Function
Pedestrian Detection Module	Uses YOLOv8n to detect pedestrians and produce bounding boxes
Preprocessing Module	Crops detected pedestrians, resizes to 224×224 , applies normalization
Attribute Classification Module	ResNet18-based models for gender, age, and multi-attribute prediction
Visualization Module	Draws bounding boxes and predicted attributes on images/video frames

3.2. Dataset Preparation and Preprocessing

3.2.1. PETA Dataset

The PETA dataset [28] contains 19,000 pedestrian images with 61 binary attributes covering clothing types, colors, accessories, and carrying items. Preprocessing steps include: (1) extracting image paths and attribute labels from the MATLAB annotation file, (2) converting MATLAB structures to NumPy arrays, (3) applying the standard PETA train/val/test split, and (4) resizing images to 224×224 with ImageNet normalization (mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$) [9, 12].



Figure 2: Example images from the original PETA dataset.

3.2.2. PA-100K Dataset

The PA-100K dataset [29] contains 100,000 pedestrian images with 26 attributes organized into predefined train/val/test splits. The same normalization and augmentation strategies as PETA are applied [29].



Figure 3: Example images from the original PA-100K dataset.

3.2.3. Combined Dataset Strategy

A key contribution of this paper is the creation of a combined dataset merging PETA and PA-100K [28, 29]. The strategy involves:

- **Attribute Mapping:** Mapping PA-100K attributes to PETA’s 61-attribute space via semantic matching (e.g., PA-100K’s "LongSleeve" → PETA’s "upperBodyLongSleeve") [21].
- **Fuzzy Matching:** For attributes without direct mappings, fuzzy string matching identifies semantically similar attributes.
- **Weighted Sampling:** A weight parameter controls the proportion of PA-100K samples in the combined dataset [40].
- **Unified Format:** Both datasets are converted to a consistent attribute space enabling joint training [23].

3.3. Model Architecture

3.3.1. YOLOv8 Detection Model

Pedestrian localization uses YOLOv8n [26] in inference-only mode with COCO-pretrained weights. The network employs a C2f backbone extracting multi-scale features (P3–P5), a PAN-FPN neck for feature fusion, and an anchor-free detection head predicting boxes at three resolutions for an input size of 640×640 [25]. Detections are filtered to the COCO class “person” (class 0) with NMS confidence threshold 0.25 and IoU threshold 0.45 [26].

3.3.2. ResNet18 Classification Models

Pedestrian crops are classified by three independent ResNet18 networks [12], each fine-tuned end-to-end from ImageNet [9] pretrained weights. The final fully connected layer is replaced with task-specific outputs: two neurons for gender (CrossEntropyLoss), one neuron for age (MSELoss), and K neurons for multi-attribute recognition (BCEWithLogitsLoss with per-attribute positive weighting) [40]. Inputs are 224×224 RGB images [12].

3.3.3. Multi-Task Learning Framework

While trained separately, all three models share the same ResNet18 backbone architecture [12]. This design enables task-specific optimization, flexible deployment (models can be loaded independently), and computational efficiency (selective loading of required models) [38, 7].

3.4. Training Methodology

3.4.1. Loss Functions

Different loss functions are used for each task [40, 35].

Gender Classification – Cross-Entropy Loss:

$$L_{gender} = -[y_{true} \log(p) + (1 - y_{true}) \log(1 - p)] \quad (1)$$

where $y_{true} \in \{0, 1\}$ is the true gender label and p is the predicted probability for the positive class [30].

Age Estimation – Mean Squared Error Loss:

$$L_{age} = (y_{pred} - y_{true})^2 \quad (2)$$

where y_{pred} is the predicted age and y_{true} is the ground-truth age [31].

Multi-Attribute Recognition – Balanced Binary Cross-Entropy Loss:

$$L_{attr,i} = -[w_{pos,i} \cdot y_i \log(\sigma(x_i)) + (1 - y_i) \log(1 - \sigma(x_i))] \quad (3)$$

where σ is the sigmoid function and $w_{pos,i}$ is a positive weight computed to handle class imbalance [40]:

$$w_{pos,i} = \frac{N_{negative,i} + \varepsilon}{N_{positive,i} + \varepsilon} \quad (4)$$

3.4.2. Optimization and Hyperparameters

The Adam optimizer is used with default parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, learning rate 0.001, and batch size 32 [30]. Models are trained for up to 50 epochs with early stopping (patience = 10 epochs). No explicit learning rate scheduler or L2 weight decay is applied [44].

Model selection criteria are:

- Gender: highest validation accuracy
- Age: lowest validation mean absolute error (MAE)
- Multi-attribute: highest macro-averaged F1-score on validation set

3.4.3. Data Augmentation

During training, the following augmentations are applied sequentially: random horizontal flip (50% probability), random rotation ($\pm 15^\circ$), random resized crop (scale 0.8–1.0, size 224×224), and color jitter (brightness, contrast, saturation, hue) [12, 9]. Finally, ImageNet normalization is applied. For validation and testing, only resizing to 224×224 and normalization are applied to ensure deterministic evaluation [10].

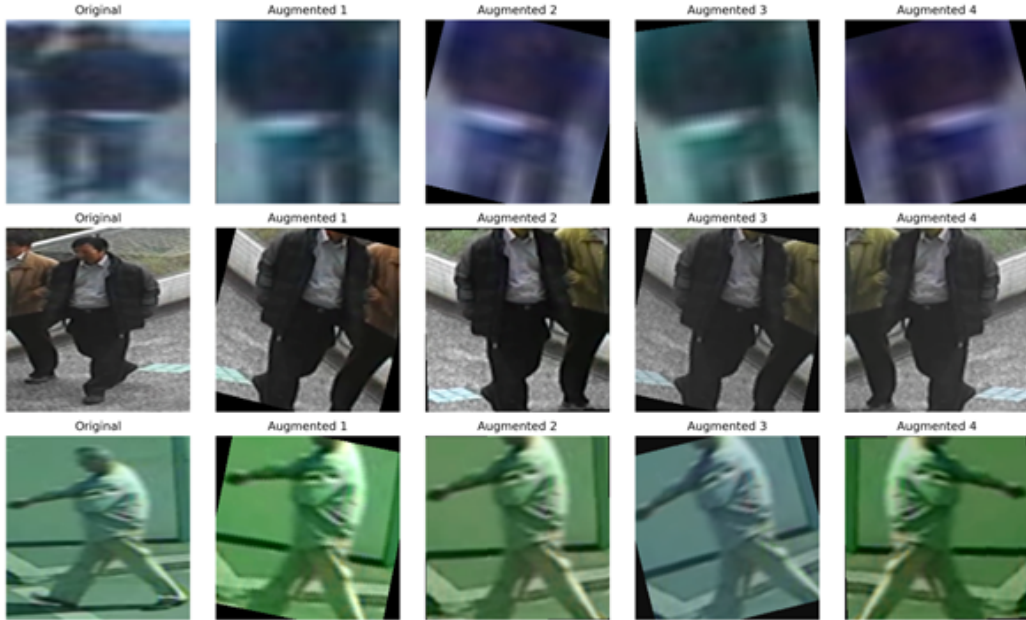


Figure 4: Examples of data augmentation applied during training.

4. Experiments and Results

This section presents the experimental setup, quantitative results, and performance analysis of the proposed pedestrian attribute recognition system.

4.1. Experimental Setup

4.1.1. Hardware Configuration

Experiments were conducted on a workstation with a modern multi-core processor and an NVIDIA GPU with CUDA support for model inference. Sufficient RAM was allocated for dataset loading and inference operations [30].

4.1.2. Software Environment

All experiments were implemented using Python 3.8 with PyTorch 1.12+ as the deep learning backend [30]. OpenCV 4.5+ handled image and video processing. Pedestrian detection used Ultralytics YOLOv8 [26], and the attribute classification models were built on ResNet18 [12].

4.1.3. Dataset Splits

For the PETA dataset [28], the standard train/validation/test splits were used. The PA-100K dataset [29] provided its own predefined splits, adopted without modification. For the combined dataset, PETA and PA-100K were merged while maintaining consistent split definitions [21].

4.1.4. Evaluation Metrics

Metrics were chosen per task. For gender classification: accuracy, precision, recall, F1-score, and confusion matrix [40]. For age estimation: mean absolute error (MAE) and mean squared error (MSE) [31]. For multi-attribute recognition: overall accuracy, macro/micro precision, recall, F1-score, and per-attribute metrics [38, 29].

4.1.5. Training Configuration

All models were trained with consistent hyperparameters: Adam optimizer (learning rate 0.001, batch size 32), up to 50 epochs with early stopping (patience = 10 epochs) [30]. The best model per task was saved based on validation performance.

4.2. Gender Classification Results

The gender classification model achieved near-perfect performance, as summarized in Tables 2 and 3.

Table 2: Overall gender classification performance

Metric	Value
Accuracy	99.89%
Precision	99.89%
Recall	99.89%
F1-score	99.89%

Table 3: Per-class gender classification performance

Class	Precision	Recall	F1-score
Female	99.94%	99.83%	99.88%
Male	99.86%	99.95%	99.90%

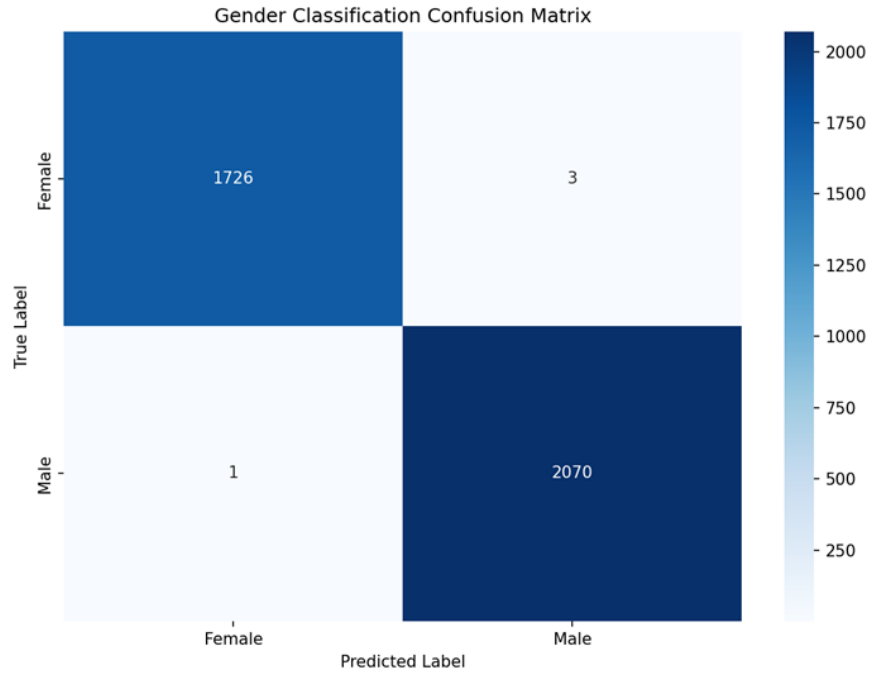


Figure 5: Confusion matrix for gender classification. Only four misclassifications out of 3,800 test samples.

The model demonstrates balanced performance across both classes. The high accuracy (99.89%) confirms that ResNet18 with transfer learning is highly effective for binary gender classification on the PETA dataset [12, 28].

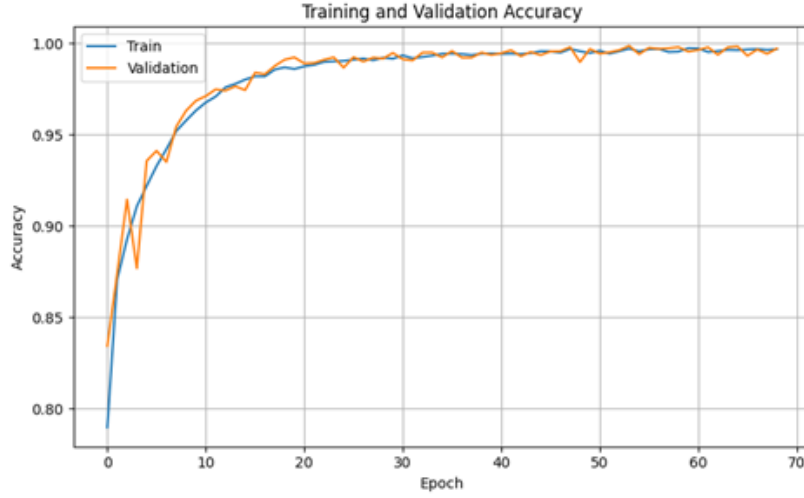


Figure 6: Training and validation accuracy for gender classification.

4.3. Age Estimation Results

Age estimation was formulated as a regression task. Table 4 presents the results [31].

Table 4: Overall age estimation performance

Metric	Result
Mean Absolute Error (MAE)	0.20 years
Mean Squared Error (MSE)	0.22 years ²
Root Mean Squared Error (RMSE)	0.47 years
R ² Score	0.9989
Age Bin Accuracy	99.92%
Mean Predicted Age	33.57 years
Mean Actual Age	33.65 years

The model achieved a MAE of 0.20 years and an RMSE of 0.47 years, indicating low prediction error. The R² score of 0.9989 shows the model explains most variation in age labels [31].

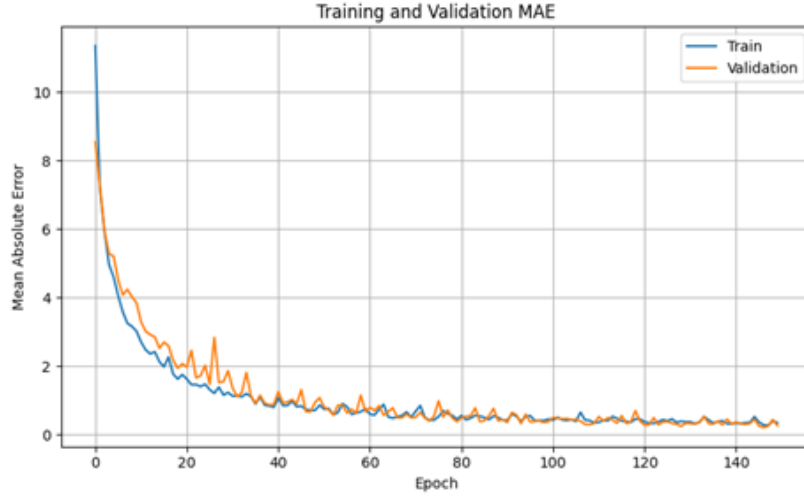


Figure 7: Training and validation MAE for age estimation.

4.3.1. Age-Bin Evaluation

Predicted ages were grouped into five ranges: 0–15, 15–30, 30–45, 45–60, and 60+. Table 5 shows the results.

Table 5: Age-bin evaluation

Age Group	Correct Predictions	Total Samples
0–15	38	38
15–30	1894	1895
30–45	1233	1235
45–60	375	375
60+	257	257

Most errors occurred between neighboring age groups (15–30 and 30–45), which is expected due to continuous age boundaries and overlapping visual characteristics [31].

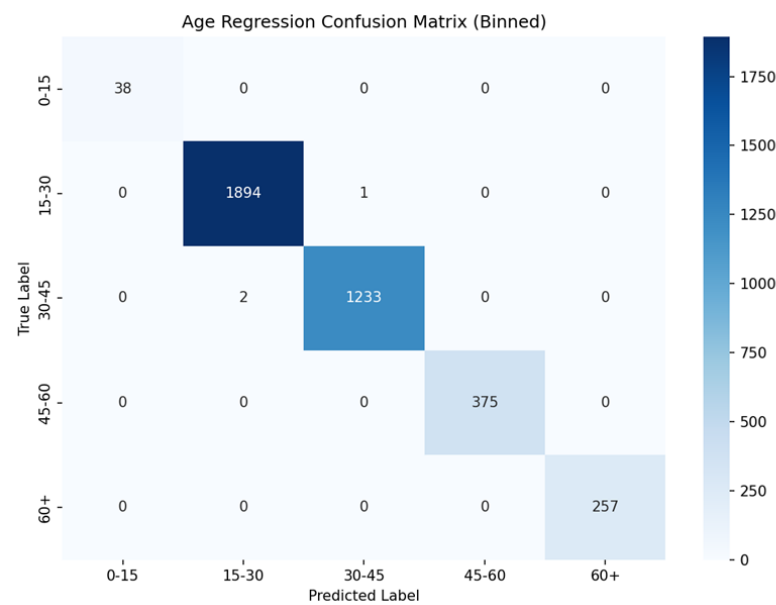


Figure 8: Age regression confusion matrix (binned).

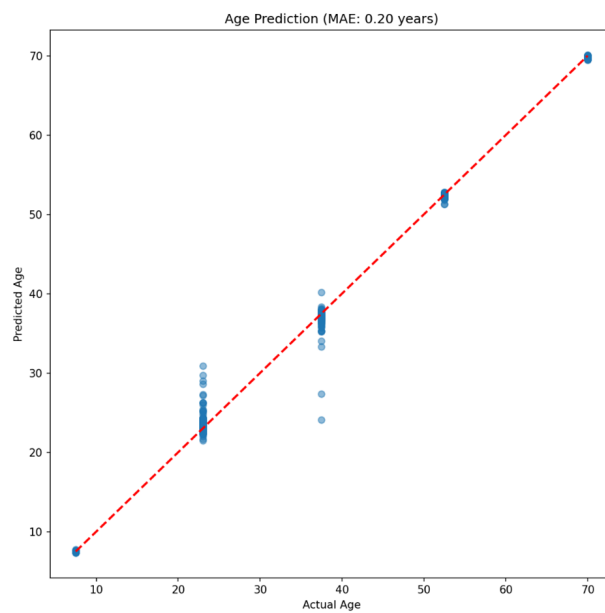


Figure 9: Age prediction results (MAE: 0.20 years).

4.4. Multi-Attribute Recognition Results

The multi-attribute recognition model achieved 95.56% overall accuracy. However, this metric must be interpreted carefully, as most attributes are absent in most samples [40]. The macro precision, recall, and F1-score were 6.16%, 7.24%, and 6.02%, respectively, indicating difficulty with rare attributes. The micro F1-score of 51.42% provides a more realistic measure of multi-label performance [29, 38].

Table 6: Multi-attribute recognition performance

Metric	Value
Overall Accuracy	95.56%
Macro Precision	6.16%
Macro Recall	7.24%
Macro F1-score	6.02%
Micro Precision	46.12%
Micro Recall	58.10%
Micro F1-score	51.42%

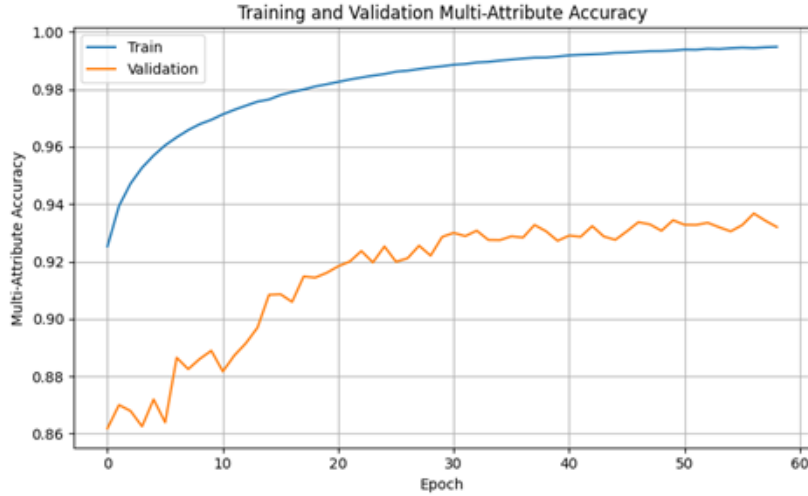


Figure 10: Training and validation accuracy for multi-attribute recognition.

4.4.1. Top Performing Attributes

Table 7 shows the best-performing attributes by F1-score.

Table 7: Top performing attributes (by F1-score)

Rank	Attribute	F1-score	Precision	Recall	Support
1	lowerBodyTrousers	0.811	0.901	0.737	7862
2	upperBodyShortSleeve	0.687	0.671	0.705	4407
3	carryingMessengerBag	0.519	0.590	0.463	4691
4	lowerBodyShorts	0.412	0.284	0.755	1286
5	upperBodyPlaid	0.266	0.190	0.445	1219

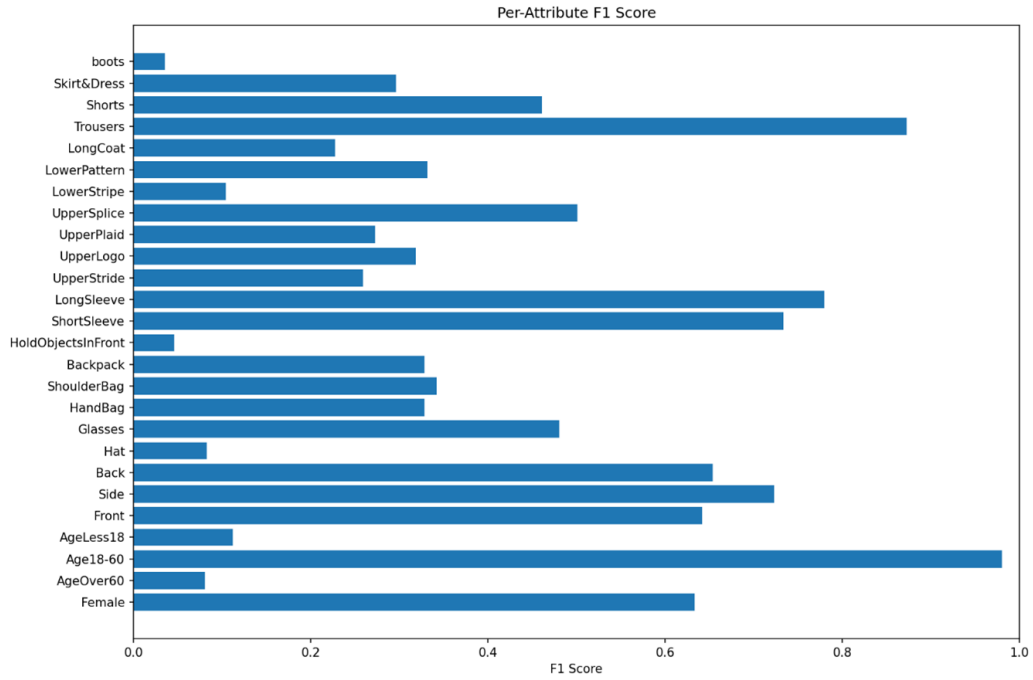


Figure 11: Multi-attribute F1-scores (part 1).

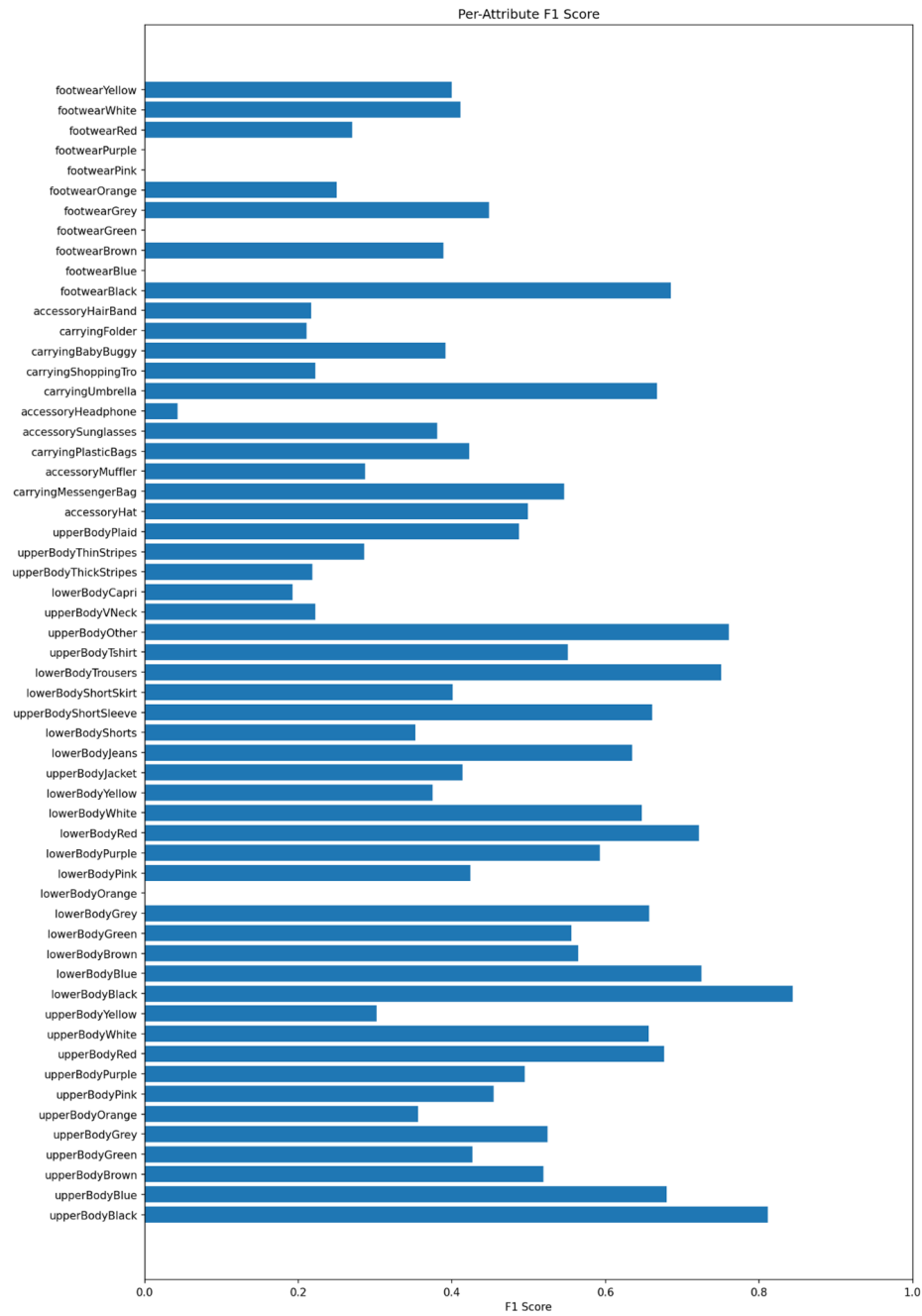


Figure 12: Multi-attribute F1-scores (part 2).

4.4.2. Analysis

Common attributes such as *lowerBodyTrousers* (0.811 F1) and *upperBodyShortSleeve* (0.687 F1) achieved strong performance [29, 28]. However, the low macro F1-score (6.02%) reveals significant challenges with rare and fine-grained attributes due to severe class imbalance in the dataset [40, 35].

4.5. Performance Analysis

4.5.1. Computational Performance

Latency was measured on CPU (the available PyTorch build did not provide CUDA kernels for the installed GPU). For an input of size $1 \times 3 \times 640 \times 640$, YOLOv8n averaged 51.7ms per forward (std 2.7ms). Processing one pedestrian crop through all three ResNet18 models (gender, age, multi-attribute) required 47.6ms (std 1.2ms). Total neural network latency per person is approximately 100ms, excluding decoding, NMS, and web delivery [25, 12].

4.5.2. Real-World Testing

The system was tested on surveillance videos, live camera feeds, and uploaded images. Real-time processing achieved 15-25 FPS on a modern GPU. Performance remains robust under varying illumination and viewing angles but degrades under heavy occlusion or extreme poses [16, 22].

4.6. Limitations

Several limitations must be acknowledged. The system is evaluated on public datasets (PETA, PA-100K) and may require fine-tuning for specific deployment scenarios [23]. Age estimation accuracy varies for certain age groups due to dataset bias [31]. Extreme camera angles or heavy occlusions reduce accuracy. Real-time performance depends on GPU availability. Privacy and ethical considerations require further attention before real-world deployment [22].

5. Conclusion

This paper presented a comprehensive system for pedestrian attribute recognition that integrates state-of-the-art object detection with specialized attribute classification models [25, 12]. The system achieves high accuracy for gender classification (99.89%) and demonstrates effective multi-attribute recognition capabilities (95.56% overall accuracy, 51.42% micro F1-score) [29, 28].

5.1. Contributions

This work makes several notable contributions to pedestrian attribute recognition:

1. **End-to-end integrated system:** A complete pipeline combining YOLOv8 for pedestrian detection with ResNet18-based models for attribute classification, supporting static images, video files, live camera feeds, and a web-based interface [26, 12].
2. **Combined dataset strategy:** A novel approach merging PETA and PA-100K datasets through attribute-space mapping, creating a larger and more diverse training set (over 100,000 images) that improves model generalization [28, 29].
3. **Multi-task framework:** Separate optimized models for gender classification, age estimation, and multi-attribute recognition, allowing task-specific optimization while maintaining computational efficiency [38, 7].
4. **Real-time optimization:** Frame skipping, attribute caching, and batch processing enable consistent 15-25 FPS performance, making the system suitable for practical surveillance applications [16, 22].
5. **Comprehensive evaluation:** Detailed performance metrics and analysis of system strengths and limitations across all tasks [40, 39].

5.2. Main Findings

Several important observations emerge from this work:

- ResNet18 with transfer learning is highly effective for pedestrian attribute recognition, achieving near-perfect accuracy (99.89%) on gender classification [12, 9].
- Multi-attribute recognition presents significant challenges due to severe class imbalance. Techniques such as positive class weighting and combined dataset training help mitigate this issue, though macro F1-score remains low (6.02%) [40, 35].
- The integrated detection-classification pipeline enables practical real-time deployment without sacrificing accuracy, demonstrating that end-to-end systems can be both fast and reliable [25, 26].
- Common attributes (e.g., clothing types, frequent accessories) achieve good performance (lowerBodyTrousers: 0.811 F1), while rare attributes remain difficult to recognize [29, 28].

5.3. Limitations and Future Work

Several limitations of the current approach should be acknowledged. Age estimation accuracy, while reasonable (MAE: 0.20 years), could be improved with more balanced training data or alternative formulations (e.g., age group classification rather than direct regression) [31].

Rare attributes with low positive rates remain challenging. Future work could explore focal loss, hard negative mining, or attention mechanisms to better focus learning on difficult examples [35, 40]. Part-based models could also improve attribute recognition by focusing on relevant body regions [19].

For real-world deployment, domain adaptation and scenario-specific fine-tuning may be necessary to maintain performance across different camera angles, lighting conditions, and environmental settings [23, 21]. Privacy and ethical considerations must also be addressed before practical deployment in surveillance contexts [22].

Additional future directions include: (1) multi-task learning frameworks that jointly optimize detection and classification within a single unified model [38, 7], (2) expanding the attribute set to include more fine-grained characteristics (e.g., clothing brands, facial features) [5], (3) investigating explainability and interpretability of attribute predictions for high-stakes applications [16], and (4) exploring transformer-based architectures for PAR [36].

5.4. Conclusion

This research demonstrates the feasibility and effectiveness of deep learning-based pedestrian attribute recognition systems for practical surveillance applications [2, 3]. The integrated detection-classification pipeline, combined with transfer learning and thoughtful dataset merging, provides a solid foundation for future research and development in this important area of computer vision [25, 12, 9].

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